Biology

Biology is the science of all life, from microscopic bacteria that cannot be seen with the naked eye to enormous animals such as elephants and whales. It includes their form and function, origin and growth, and evolution and distribution.

NEEDED FOR LIFE

All life forms need the same essentials to survive. Few forms of life can exist without most of these basic necessities.

ALL ORGANISMS NEED THE SAME BASIC ESSENTIALS IN ORDER TO LIVE

WATER
All living things are made of cells, which need water to exist – most life forms are mainly made up of water.

ENERGY SOURCE
Life forms need energy to grow and move around. Plants use sunlight to make energy. Animals get energy by eating plants or each other.

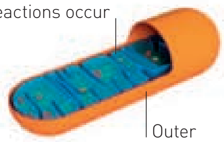
OXYGEN
Oxygen in air or water is necessary for all life.

ESSENTIAL CHEMICALS
The chemicals hydrogen, nitrogen, and carbon are essential for life. Plants get them from soil, while animals absorb them from food.

THE RIGHT TEMPERATURE
Few living things can exist in extremely hot or cold temperatures.

WHAT IS A CELL?

Cells are the building blocks of life. The cells of all living things except archaea and bacteria contain a nucleus, mitochondria, and other organelles. Cells can be specialized to perform different functions – for example, we have nerve, muscle, and bone cells. The human body has around 75 trillion cells, whereas less complex organisms may have only one.



Inner membrane where chemical reactions occur

Outer membrane

MITOCHONDRION
This is the part of the cell that releases energy from food molecules within the body.

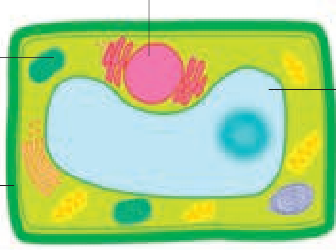
Vacuoles store nutrients or waste

Ribosomes are the protein builders of the cell

ANIMAL CELL
An animal cell contains lots of "machines" called organelles that perform special jobs (such as the mitochondrion).

A jelly-like fluid called cytoplasm fills the space between the organelles

The nucleus is the cell's control centre. It sends chemical instructions to other parts of the cell



Nucleus

Chloroplasts convert sunlight into energy

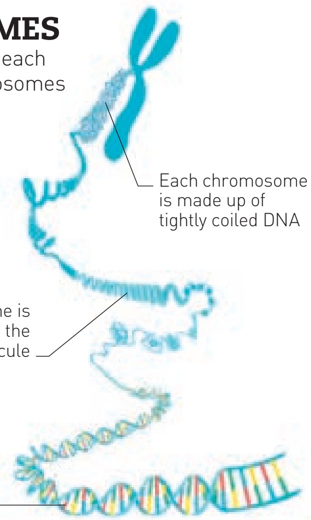
Large vacuole filled with cell sap

Rigid cell wall

PLANT CELL
These have much in common with animal cells, but they also have rigid cell walls and chloroplasts.

CHROMOSOMES

Within the nucleus of each cell there are chromosomes that carry DNA. DNA contains genes that determine how an organism looks and functions. Humans have 46 chromosomes (23 pairs).



Each chromosome is made up of tightly coiled DNA

Each gene is a section of the DNA molecule

DNA is a long molecule arranged in a double-helix shape

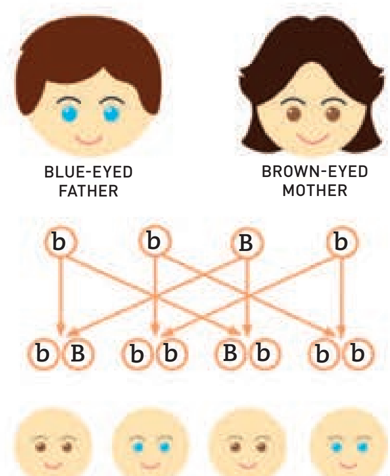
GENES

Our genes are inherited from our parents – half from mum and half from dad – and they dictate things like eye colour. Each person has two versions of each gene, called alleles, which together make up their genotype. One allele is often dominant over another, which means that that feature is the one seen in the person.

KEY

b The recessive allele. A child must have two b alleles to have blue eyes.

B The dominant allele – a child with one or two B alleles will have brown eyes.



Each parent carries a different combination of genes for eye colour

Parents' genes

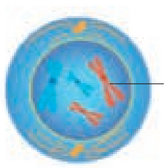
Possible gene combinations of children

Children's possible eye colours

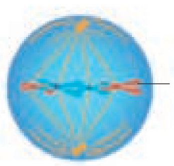
GENETICS IN ACTION
The mother here has one recessive and one dominant allele. The father has two recessive alleles. This means it is equally likely that they have a brown- or blue-eyed child.

CELL DIVISION

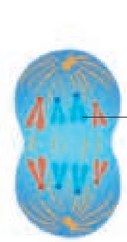
Organisms develop from a single cell, which divides again and again. Over the organism's lifetime, its cells are continually replaced in a process called mitosis.



Chromosomes



Chromosomes duplicate



Doubled chromosomes are pulled apart

Cell splits into two daughter cells, each with a full set of chromosomes

1 FIRST STAGE
The cell contains chromosomes that can be copied to make new identical chromosomes.

2 CHROMOSOMES ALIGN AND COPY
The wall of the nucleus breaks down and chromosomes line up in the middle of the cell.

3 SEPARATION
The doubled chromosome is pulled in half, so one chromosome moves to each end of the cell.

4 "DAUGHTER" CELLS FORM
Cell splits into two identical cells and the nuclear wall reforms.